PD Group:	Name:

ANDERSON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS

BIOLOGY
Higher 2

9648/02

12 SEPT 2011 MONDAY

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graph or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Paper 2

Section A

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

All working for numerical answers must be shown.

Section B

Answer any **one** question.

Write your answers on the separate writing paper provided.

A **NIL** return is required.

At the end of the examination, fasten your essay answers to the answer booklet.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
Total	

The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **19** printed pages.

SECTION A

Answer **all** the questions in this section.

1 Fats can be transported in an aqueous medium such as blood by complexing with proteins called apoproteins. This forms a spherical complex known as lipoproteins. The lipoprotein usually consists of other molecules such as phospholipid, cholesterol and a variety of other lipids in its core as shown in Fig. 1.1.

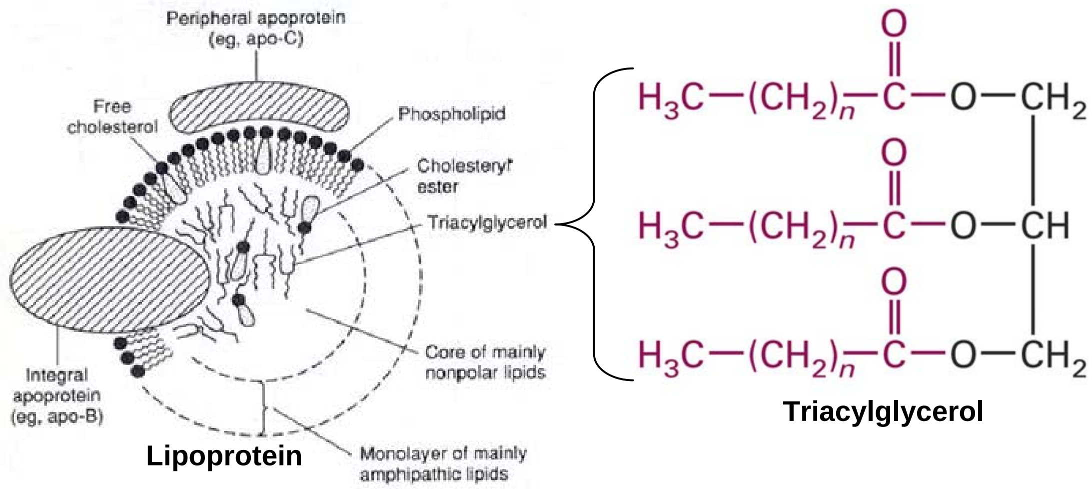

Fig. 1.1

(a) With reference to Fig. 1.1,

(i) State two ways in which the structural properties of apo-B differ from apo-C.

1……

……

2……

…… [2]

(ii) Explain why a triacylglycerol is a good energy storage molecule in mammals.

……

……

…… [2]

(b) Suggest why lipoproteins, as shown in Fig. 1.1, are normally spherical in shape.

……

……

…… [2]

(c) Chitin is a polymer of N-acetylglucosamine. It is the main component of fungal cell walls. The structure of chitin and where they are found on the fungal cell wall is shown in Fig. 1.2.

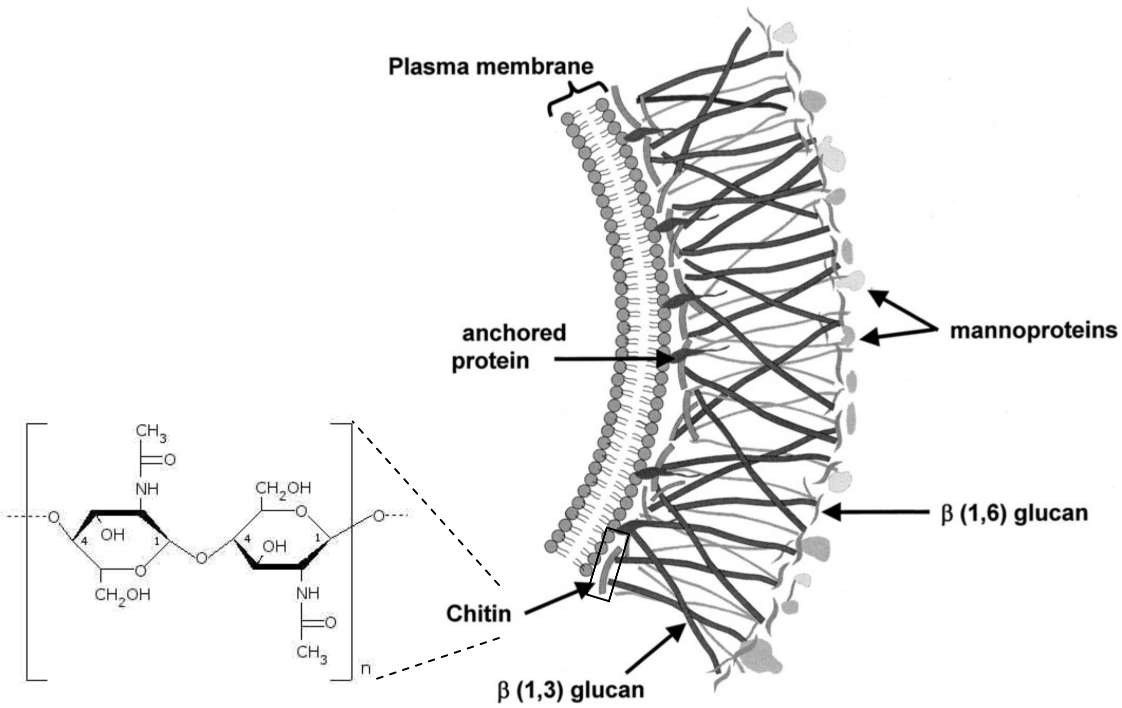

Fig. 1.2

With reference to Fig. 1.2,

(i) describe two similarities in the molecular structure between chitin and cellulose.

1……………………………………………………………………………………………………

……………………………………………………………………………………………………

2……………………………………………………………………………………………………

…………………………………………………………………………………………………… [2]

(ii) suggest why the fungal cell wall is tough.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

…………………………………………………………………………………………………… [2]

[Total : 10]

2 Fig 2.1 shows a molecule of tRNA and the enzyme that attaches the correct amino acid to it.

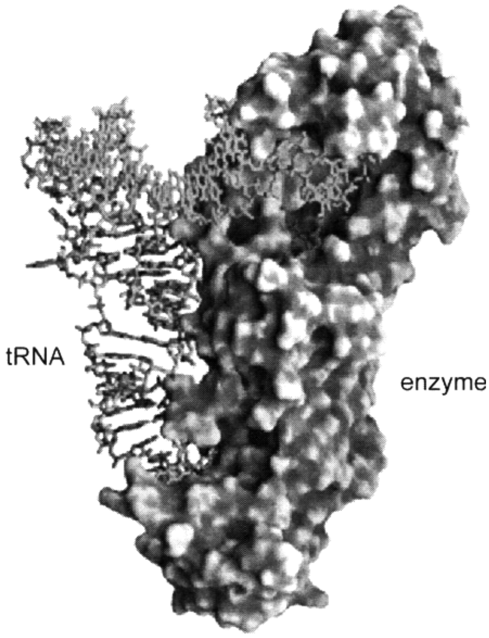

Fig. 2.1

(a) With reference to Fig. 2.1, explain how the enzyme is suited to its function.

………..

………...

………..

………... [3]

(b) The average molecular weight of proteins encoded in the human genome is about 50,000 daltons. Given the average molecular mass of an amino acid is about 110 daltons and 5% of the initial transcript is converted to mature mRNA, estimate how long it will take a muscle cell to transcribe a gene for an average protein. Assume that the transcription rate is 20 nucleotides per second.

Show your working and express your answer to the nearest whole number (in minutes).

Answer: …………………… minutes [2]

(c) Edeine is an antibiotic that inhibits protein synthesis but has no effect on either DNA synthesis or RNA synthesis. When added to the cell extract of an immature red blood cell, edeine stops protein synthesis after a short lag, as shown in Fig. 2.2. By contrast, cycloheximide, which is also an inhibitor, stops protein synthesis immediately. Protein synthesis is measured via radioactivity in haemoglobin.

Analysis of the edeine-inhibited cell extract showed that no polyribosomes remained at the time protein synthesis had stopped. Instead, all the globin mRNA were found associated with small ribosomal subunit and initiator tRNA.

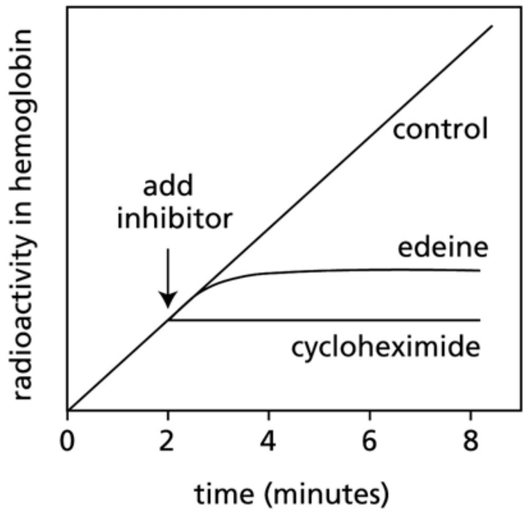

Fig. 2.2

(i) Explain how protein synthesis is measured via radioactivity in haemoglobin.

………

……… [1]

(ii) Describe how edeine inhibits protein synthesis.

………

………

……… [2]

(iii) Explain why there is a lag between the addition of edeine and cessation of protein synthesis.

………

………

……… [2]

[Total : 10]

3 Bacteria can undergo genetic recombination, a process by which genetic information from one bacterium is transferred to, and then recombined with, that of another bacterium.

Experiments have shown that 2 strains of bacteria can undergo conjugation when grown in a common medium. However, when they are separated by a filter in the Davis U-tube, as shown in Fig. 3.1, conjugation cannot occur.

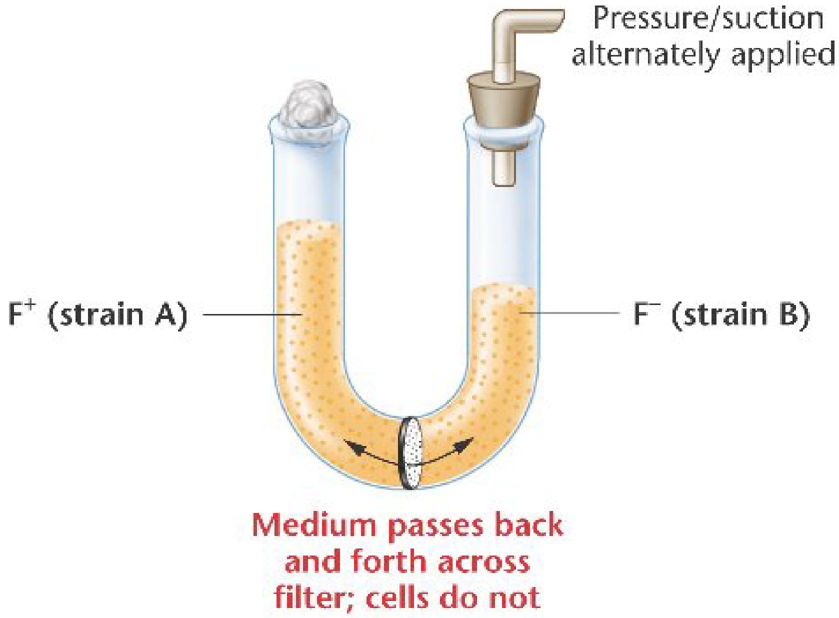

Fig. 3.1

(a) With reference to Fig. 3.1, explain why conjugation cannot occur when the 2 strains of bacteria are placed in the Davis U-tube.

……

……

……

……

…… [3]

(b) In another experiment investigating possible recombination in the bacterium *Salmonella typhimurium*, when researchers mixed *Salmonella* strains X and Y in a common medium, they recovered recombinants.

In a separate experiment using the Davis U-tube, the strains X and Y were separated by a filter, thus preventing cell contact but allowing growth to occur in a common medium. Surprisingly, when samples were removed from both sides of the filter, recombinants were recovered only from the side of the tube containing strain X bacteria. Researchers postulated that a filterable agent was released by the strain Y cells and responsible for transferring the new genetic information.

Three subsequent observations were useful in identifying the filterable agent:

1. The filterable agent was released by the strain Y cells only when they were grown in association with strain X cells.
2. The addition of DNase, which enzymatically digests naked DNA, did not render the filterable agent ineffective.
3. The filterable agent could not pass across the filter of the Davis U-tube when the pore size was reduced below the size of bacteriophages.

(i) Suggest the source of the new genetic information which strain X cells received.

…………………………………………………………………………………………………… [1]

(ii) State the process of genetic recombination that has occurred.

…………………………………………………………………………………………………… [1]

(iii) Describe how the filterable agent could have transferred the new genetic information to strain X.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

…………………………………………………………………………………………………… [3]

(iv) Suggest how strain Y could have produced the filterable agent.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

…………………………………………………………………………………………………… [2]

[Total : 10]

4 Iron ion concentration can be kept constant by ferritin proteins and transferrin receptor proteins.

At high intracellular iron ion concentration, ferritin proteins will bind and store excess iron ions, while transferrin receptor proteins are inactive. At low intracellular iron ion concentration, ferritin proteins are inactive, while transferrin receptor proteins will transfer and increase iron ions in the cytoplasm.

Production of ferritin protein and transferrin receptors are regulated at the mRNA level by active IRE-binding proteins (IRPs) which bind to an iron response element (IRE) on the 5' untranslated region (5' UTR) of ferritin mRNA as shown in Fig. 4.1, and at the 3' untranslated region (3' UTR) of transferrin mRNA as shown in Fig. 4.2 respectively.

Ferritin proteins

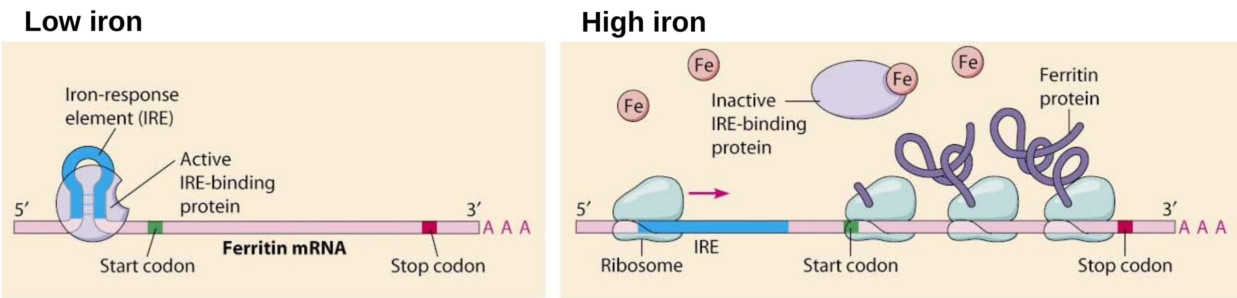

Fig. 4.1

Transferrin receptor proteins

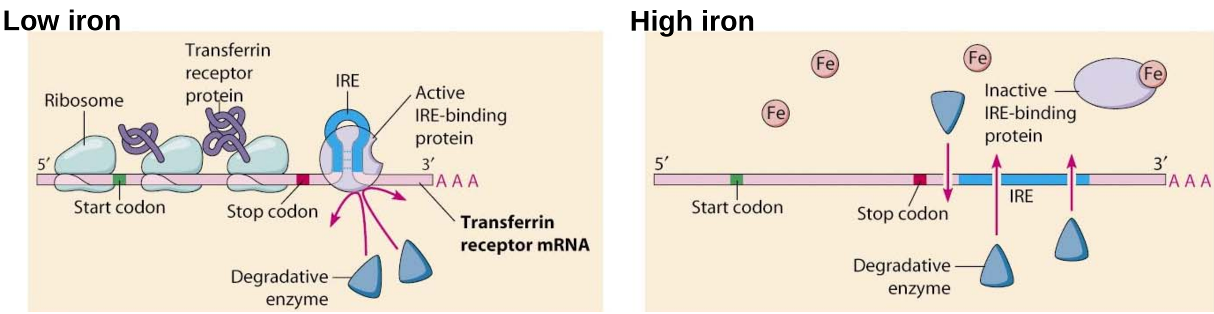

Fig. 4.2

(a) With reference to the structures of ferritin mRNA in Fig. 4.1 and transferrin receptor mRNA in Fig. 4.2,

(i) explain how the active IRE-binding protein can bind to IRE.

………………………………………………………………………………………………………

………………………………………………………………………………………………………

……………………………………………………………………………………………………… [2]

(ii) describe how expression of the ferritin protein and transferrin receptor genes may be controlled at the translational level when iron ion concentration is **low**.

………

………

………

………

………

……… [5]

(**b)** Suggest the importance of the poly(A) tail to the ferritin protein and transferrin receptor mRNA in translation control.

………

………

……… [2]

(c) Suggest why the degradative enzymes can cut the transferrin receptor mRNA as shown in Fig. 4.2, but not the ferritin protein mRNA.

………

………

……… [2]

(d) In general, mature mRNAs have low stability. Suggest how cells ensure maximum protein synthesis during the life-time of these mRNAs.

………

……… [1]

[Total : 12]

5 Fig. 5.1 shows how a gene can determine the eye colour in *Drosophila Melanogaster*, commonly known as the fruit fly.

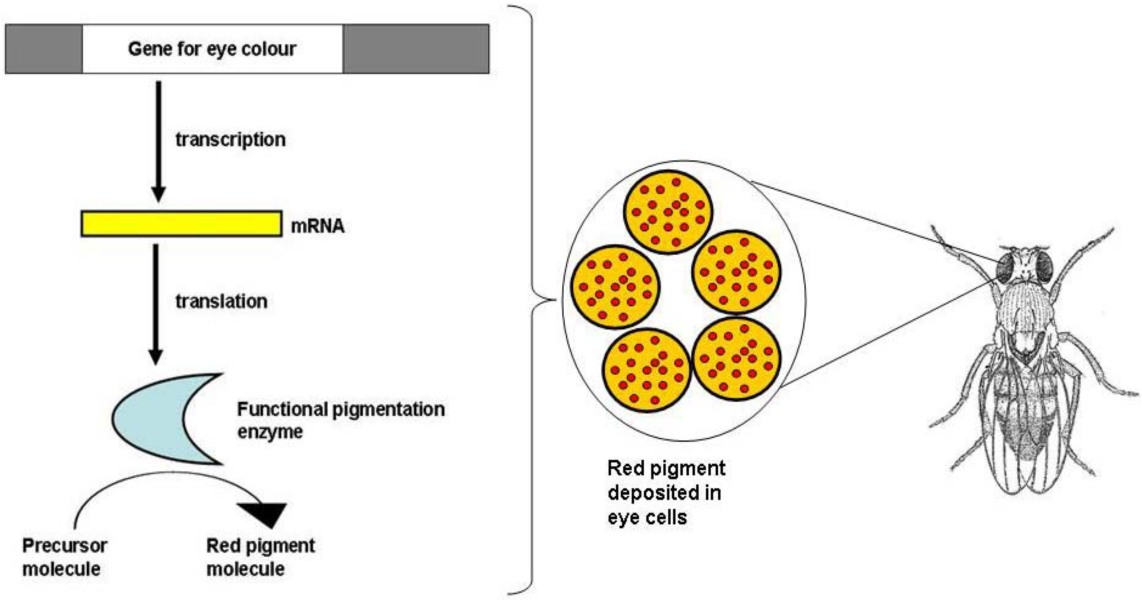

Fig. 5.1

(a) **(i)** With reference to Fig. 5.1, explain how a mutant allele of the gene coding for eye colour can result in the fruit fly having white eyes.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

…………………………………………………………………………………………………… [2]

(ii) Suggest why mutant white-eyed flies are rare in natural populations.

……………………………………………………………………………………………………

…………………………………………………………………………………………………… [1]

(b) In the early 20th century, Thomas H. Morgan provided evidence to prove unambiguously that genes were situated on chromosomes when he discovered a particular eye colour mutation in the fruit fly.

In one experiment, he crossed a red-eyed female fly with a white-eyed male fly and all the F_1 generation flies had red eyes. From the F_1 generation, when a red-eyed female is crossed with a red-eyed male, 2144 flies were produced in the F_2 generation. There were 1055 female flies and they all had red eyes. Out of the remaining male flies, 542 of them had white eyes and the rest had red eyes.

From this experiment, he was able to conclude that the mutant allele for eye colour was inherited in a recessive manner along with the X chromosome in fruit flies.

(i) Draw a genetic diagram in the space below to show the cross described. [4]

(ii) Explain how it could be deduced from the results of the cross that the mode of inheritance for this trait was X-linked recessive.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

…………………………………………………………………………………………………… [2]

(c) The observed numbers are usually different from the expected numbers in any genetic cross.

(i) Suggest two reasons why such a difference may occur in a monohybrid cross, referring only to events after meiosis.

1……………………………………………………………………………………………………

……………………………………………………………………………………………………

2……………………………………………………………………………………………………

…………………………………………………………………………………………………… [2]

(ii) Outline how you could test whether the observed results differ significantly from the expected results (Formulae not required).

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

…………………………………………………………………………………………………… [2]

[Total : 13]

6 Fig. 6.1 is an electron micrograph showing the synapse at an axodendritic junction between a pre-synaptic neurone and a post-synaptic neurone.

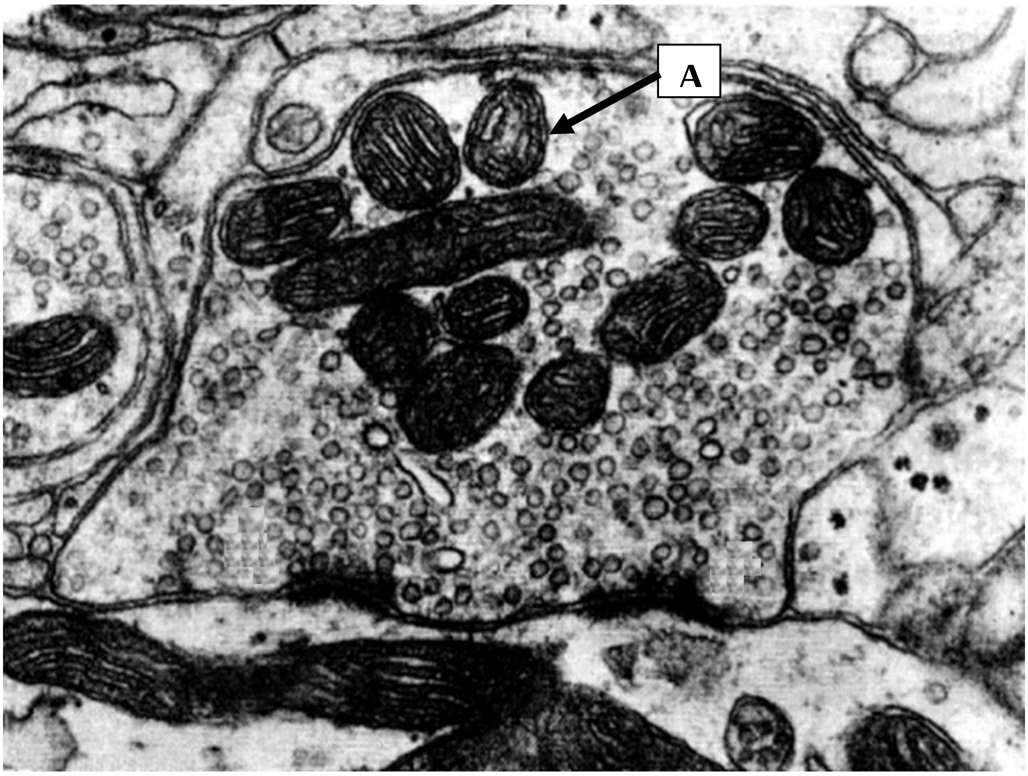

Fig. 6.1

(a) **(i)** Identify organelle **A**.

……………………………………………………………………………………………… [1]

(ii) Explain why a large number of organelle **A** is present in the synaptic knob.

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

……………………………………………………………………………………………… [3]

(b) Explain why nerve impulses can only travel in one direction across the synapse.

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

……………………………………………………………………………………………… [3]

(c) Opioids are a large group of chemical compounds that can bind to opioid receptors found on synaptic membranes of presynaptic neurones in the body. Generally, opioids decrease perception of pain by preventing an action potential from being transmitted across a synapse.

Fig. 6.2 shows a possible cell signaling pathway through which an opioid can exert its effect. The diagram is not drawn to scale.

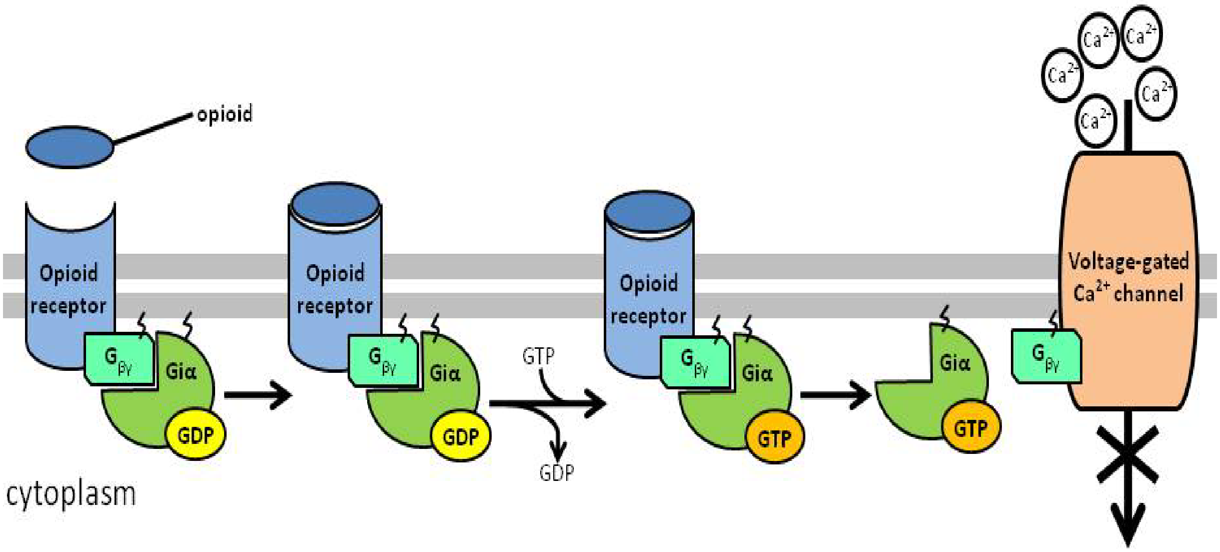

Fig. 6.2

With reference to Fig. 6.2, explain how the binding of the opioid to its receptor will prevent an action potential from being transmitted across a synapse.

………

………

………

………

………

……… [4]

(d) A few classes of endogenous opioids, such as enkaphalins, endorphins and dynorphins, are synthesized in the body and function as neurotransmitters. Endogenous opioids are short peptides made up of a few amino acids. Fig. 6.3 shows the amino acid sequences of several endogenous opioids.

endogenous opioid	amino acid sequence
[Met]enkephalin	Tyr-Gly-Gly-Phe-Met
[Leu]enkephalin	Tyr-Gly-Gly-Phe-Leu
Dynorphin A	Tyr-Gly-Gly-Phe-Leu-Arg-Arg-Ile
α-Neoendorphin	Tyr-Gly-Gly-Phe-Leu-Arg-Lys-Tyr-Pro-Lys
β-Neoendorphin	Tyr-Gly-Gly-Phe-Leu-Arg-Lys-Tyr-Pro

Fig. 6.3

Suggest why the sequences of the first four amino acids of these endogenous opioids are the same.

………..

………...

……….. [2]

[Total : 13]

7 Fig. 7.1 shows the distribution of the giant tortoises across the various islands of the Galápagos archipelago. Some sub-species are distributed across the southern part of the largest island named Isabela. In addition, three other species, *G. chilensis*, *G. carbonaria*, and *G. denticulate* are also found on the west coast of Ecuador, a country in the South America continent.

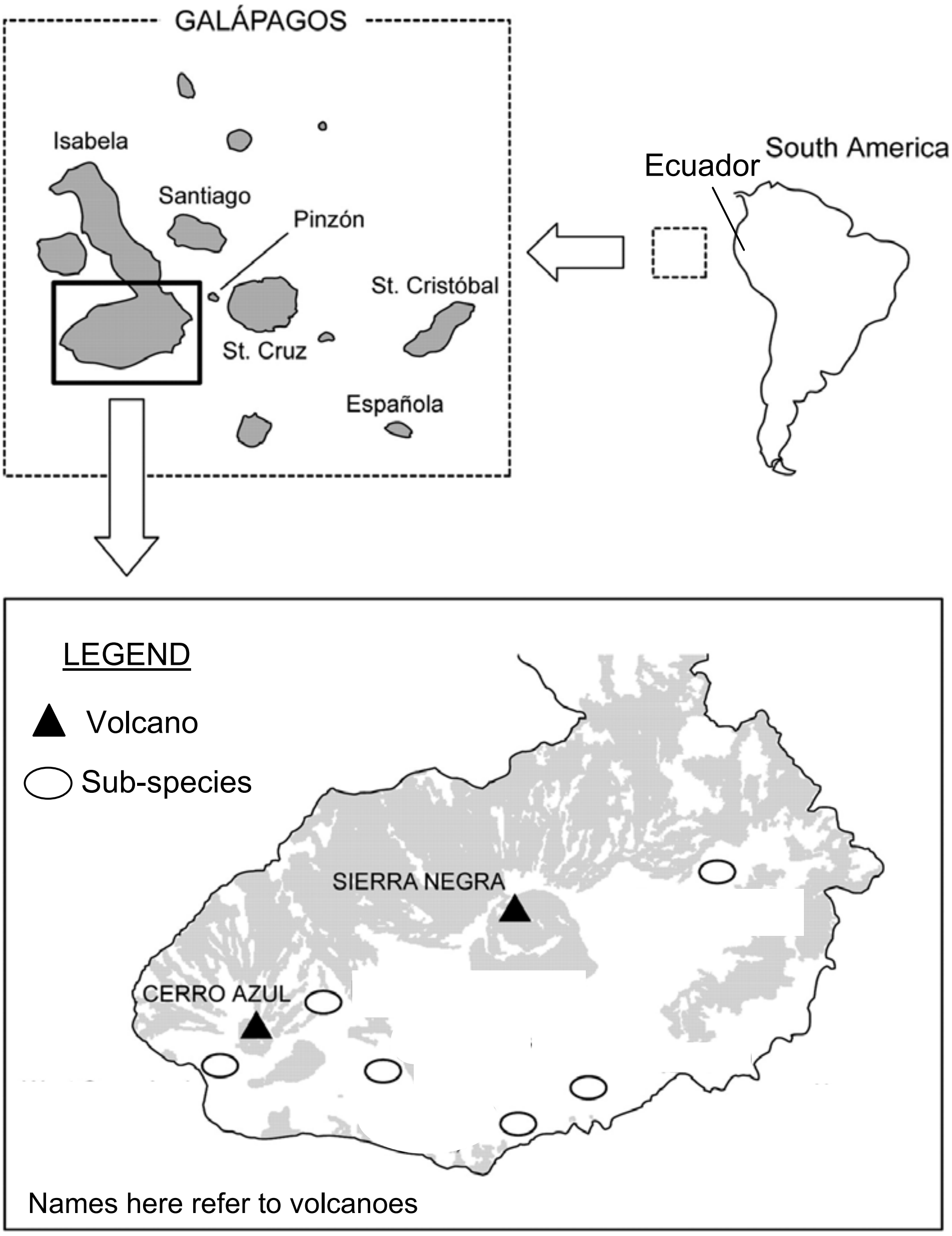

Fig. 7.1

The giant tortoises had been previously classified based on morphological differences, primarily using carapace (shell) shape, which varies from domed to saddleback with intermediate forms also occurring.

(a) **(i)** With reference to Fig. 7.1, explain how biogeography supports the evolutionary deductions that the tortoises are all evolutionally related.

………

………

……… [2]

(ii) Suggest why the previous classification using carapace shape may not be an accurate way of determining the evolutionary relationships accurately.

………

………

……… [2]

(iii) Suggest why the sub-species found on the southern part of Isabela island did not evolve to become distinct species.

………

………

……… [2]

(b) To further understand the phylogeny of the giant tortoises, the 16S rRNA and mitochondrial DNA (mtDNA) of the different populations were analyzed for their molecular homology. A phylogenetic tree is then constructed, and three pictures of giant tortoises with different carapace shapes are shown in Fig. 7.2. *G. chilensis*, *G. carbonaria*, and *G. denticulate* are also represented in the phylogenetic tree.

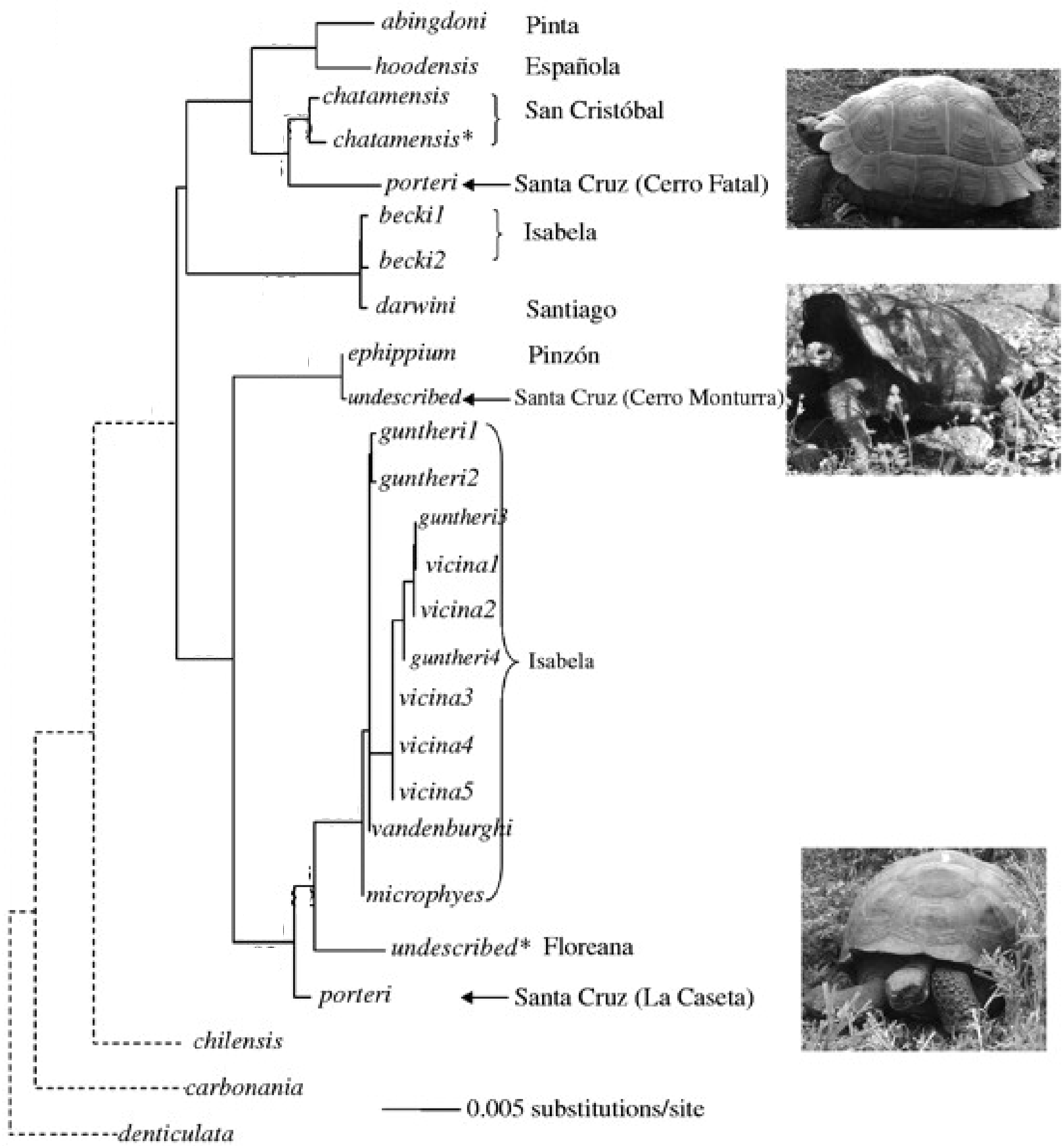

Fig. 7.2

(i) Suggest two reasons why 16S rRNA is used to construct the phylogenetic tree.

………

………

……… [2]

(ii) With reference to both Fig. 7.1 and Fig. 7.2, explain how the phylogenetic tree can support Darwin's theory of natural selection.

………

………

………

………

………

……… [4]

[Total : 12]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in section (a), (b) etc., as indicated in the question.

8 **(a)** Explain how fluidity of biological membranes can be maintained and the importance of fluidity to membrane function. [8]

(b) Describe the roles of membranes in photosynthesis and in respiration. [8]

(c) Describe the roles of membranes in the reproductive cycle of a named virus. [4]

9 **(a)** Describe the role of meiosis in natural selection. [4]

(b) Explain how genetic variation may be preserved in a natural population. [8]

(c) Describe how the differences in the structure and organization of genetic material in eukaryotes and prokaryotes affects the way genes are expressed. [8]